



New Heights Old Traditions

TAMILNADU COLLEGE OF ENGINEERING

Accredited by NAAC with Grade 'A'

(Approved by AICTE & Affiliated to Anna University, Chennai)



Department of Computer Science and Engineering

Project Second Review

DATE : 05 . 03 . 2026

Session Based – Cloud Integrated File Transferring Ecosystem

Project members:

Abineeth T M : 714222104004

Basil V Mathew : 714222104008

Inbarasu M : 714222104021

Koushick G V : 714222104029

Guided by:

Dr. Rajesh Babu
Head of department / CSE

Introduction

In the modern digital era, secure and efficient sharing of confidential documents has become a critical requirement for individuals, organizations, and enterprises. Traditional file-sharing methods often lack sufficient security measures, making sensitive information vulnerable to unauthorized access, data breaches, and cyber-attacks. To address these challenges, the Session Integrated Confidential Document Sharing Application is designed to provide a secure, reliable, and user-friendly platform for sharing sensitive documents. The system integrates session-based authentication and encryption techniques to ensure that only authorized users can access shared files within a limited time frame. This application enhances data confidentiality, integrity, and secure transmission, making it suitable for professional, academic, and business environments.

Abstract

The Session Integrated Confidential Document Sharing Application is a secure file-sharing platform that enables users to transfer sensitive documents safely using session-based authentication and encryption techniques. The system establishes a secure session between the sender and receiver, ensuring that documents are accessible only during the active session. Once the session expires, the shared files become inaccessible, thereby preventing unauthorized reuse or leakage. The application incorporates features such as encrypted data transfer, real-time device discovery, access control mechanisms, and secure storage handling. By combining these technologies, the system provides a robust solution for confidential document sharing, reducing security risks and improving trust in digital communication systems.

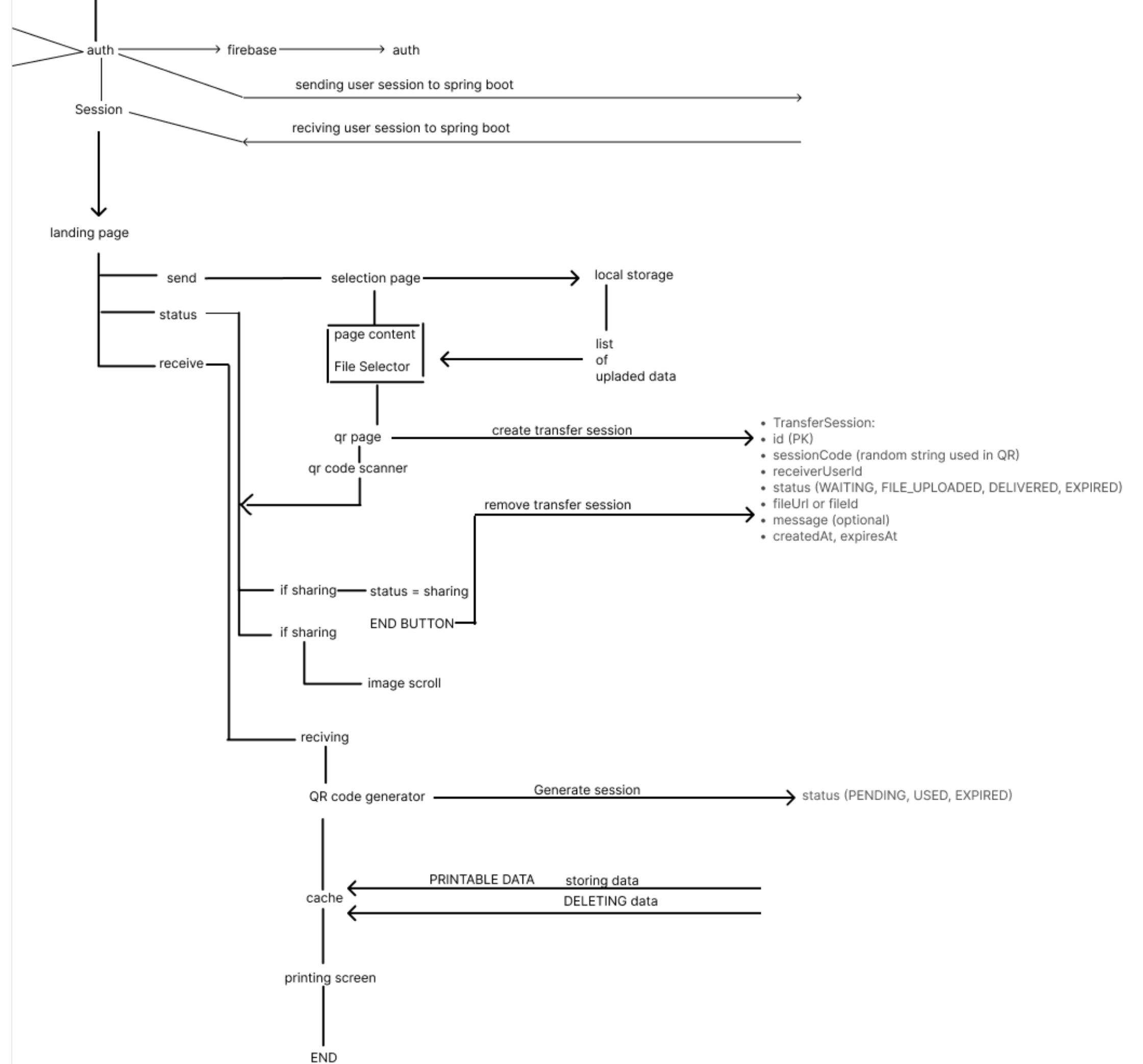
Literature Survey

Author & Year	IEEE Paper Title	Methodology / Tech
Zhang, Y. & Lee, W. (2019)	Secure Peer-to-Peer File Sharing Using End-to-End Encryption	End-to-End AES Encryption RSA Key Exchange Peer-to-Peer Communication Model
Kumar, S. & Patel, R. (2020)	Session-Based Authentication for Secure Data Transmission	Session Tokens Time-Limited Access Control HTTPS Secure Communication
Chen, L. et al. (2021)	Secure Mobile File Sharing Using WiFi Direct	WiFi Direct Peer Discovery Encrypted Socket Communication Android-Based Implementation

Proposed Method

- Session based file transfer
- Cloud integration
- User authentication
- Data set shared via QR
- Session maintained by multilevel security

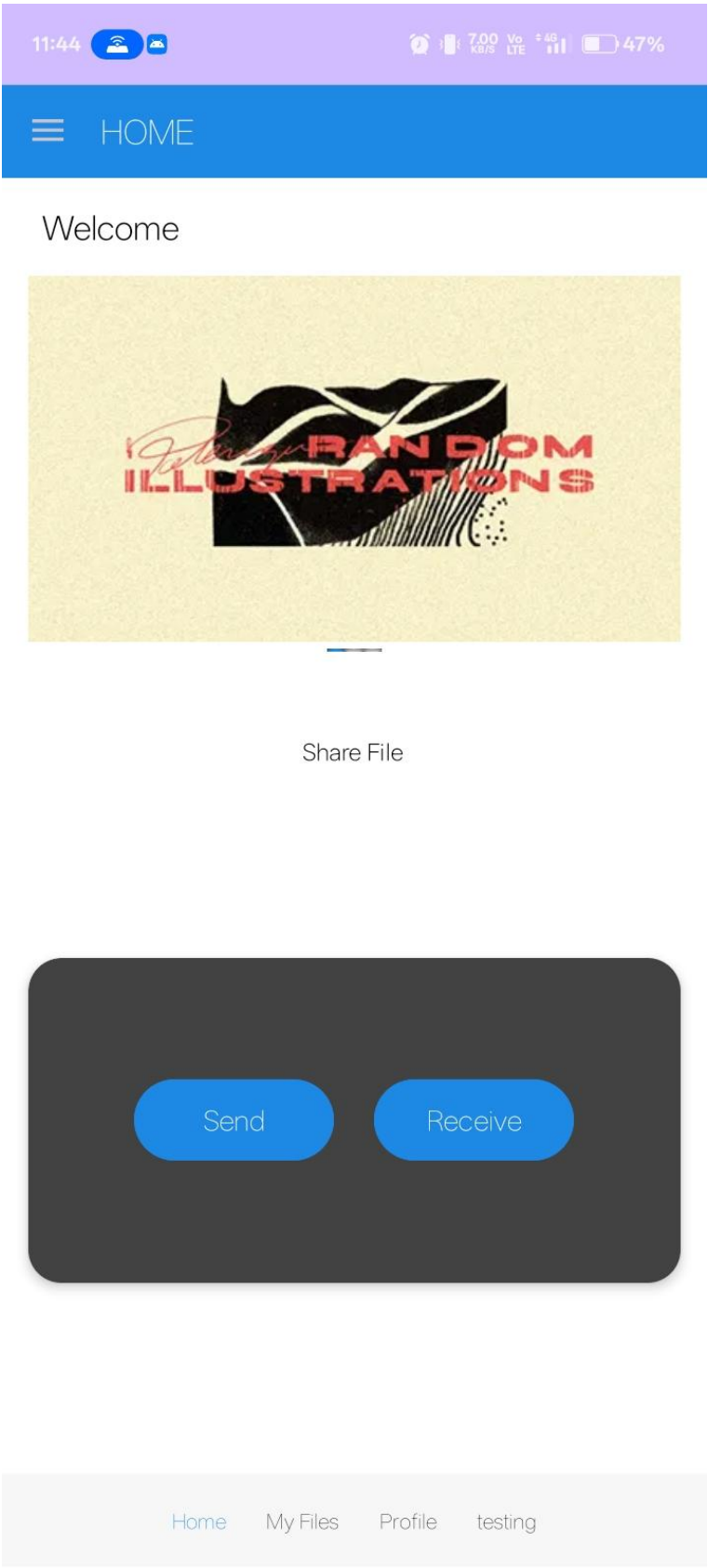
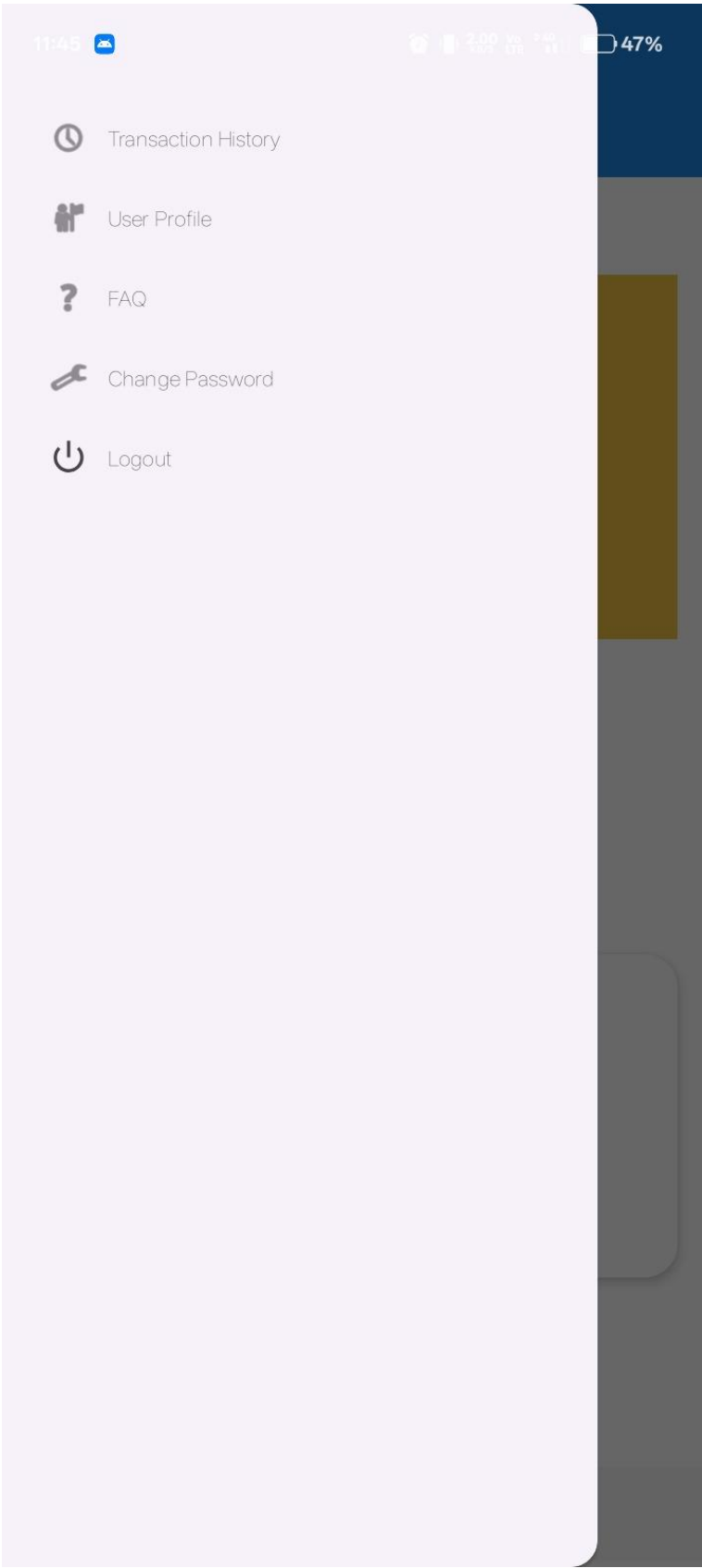
Flow Chart



Tools And Methodology Used

- **SpringBoot** : **Backend and api management**
- **Android** : **Main application interface**
- **Render** : **Backend hosting**
- **Supabase** : **Datamanagement**
- **Android Studio**: **Android app development**
- **Spring tools** : **Backend development**

Simulation And Implementation



References

- 1) William Stallings, Cryptography and Network Security,
- 2) Pearson Education.Andrew S. Tanenbaum, Computer Networks,
- 3) Pearson Education.Android Developers Official Documentation – WiFi Direct and Bluetooth APIs.OkHttp Official Documentation.
- 4) IEEE Research Papers on Secure File Sharing and Authentication.OWASP Mobile Security Guidelines.

Thank you